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RESEARCH ARTICLE

Time Series-Based Fault Detection and Classification in IEEE 9-Bus Transmission Lines Using Deep Learning

SOMCHAT JIRIWIBHAKORN^{ID}, (Senior Member, IEEE), AND SHAZIA KANWAL^{ID}

School of Engineering, King Mongkut's Institute of Technology Ladkrabang (KMITL), Bangkok 10520, Thailand

Corresponding author: Somchat Jiriwibhakorn (somchat.ji@kmitl.ac.th)

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ABSTRACT Transmission line faults present a significant threat to the stability of power systems, potentially causing widespread outages. Timely detection of these faults is essential to prevent substantial disruptions in the power supply. This paper explores a time series-based deep learning technique for fault detection and classification in the IEEE 9-bus system. Post asymmetrical fault current and voltage time series data have been used to train a convolutional neural network (CNN), representing normal and faulty conditions, with convolutional and ReLU layers. A fully connected layer is used to detect features without missing critical information of the signal, achieving MSE as zero for fault detection and 0.0149 for fault classification. This demonstrates the effectiveness of CNNs for real-time fault detection and classification in complex power grids. The robustness of the CNN model indicates its potential for deployment in practical applications, enhancing the reliability and resilience of the transmission network. Using deep learning techniques opens opportunities for further improvements in fault detection and location strategies within the power grid.

INDEX TERMS Convolutional neural network, transmission line, asymmetrical faults, IEEE 9-bus system.

I. INTRODUCTION

Fault detection and classification are critical processes in the transmission lines of today's complex grid system. Timely detection of fault and classification is a major area of research to discover and develop a relaying system for this critical task to control and save the power system for the power engineers and operators of utilities. So that the flow of power is not disturbed and hence the economic conditions of the countries. Transmission lines are the backbone of power systems. They not only carry an enormous amount of power from the generating station to the distribution system but also connect different parts of the whole network for the smooth operation of the power [1]. There are many traditional and modern techniques for detecting faults. Accurate transmission line fault recognition is crucial for protection systems to prevent mishaps. Existing algorithms have limited accuracy

in complex electrical power systems, where switching operations may be mistaken for faults. In some fault detection algorithms, the accuracy is higher; however, the algorithms are complicated to understand the prediction it makes [2]. The researchers are working towards the goal of precisely detecting the fault with higher accuracy, even in complex situations.

In modern methods, Artificial intelligence (AI) based methods such as artificial neural networks, adaptive neuro-fuzzy inference systems, and deep learning methods [3], [4] have been introduced for precisely detecting and classifying the fault. It is used for fault detection. The artificial neural networks-based methodology was used for fault detection and location, with a minimum error percentage for different fault locations [5]. Deep learning has improved performance in various applications; unsupervised based feature learning from fault signal has been introduced for fault detection and classification, using convolutional auto sparse in [6] achieving high accuracy and robustness; however, the

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convolutional sparse auto encounter may encounter computational complexity and deployment challenges. Deep learning has wide application in fault detection [7] and introduces a battery data trust framework using a CNN-based model to detect and classify false sensor and communication data in battery energy storage systems (BESSs). This approach enhances the safety and reliability of BESS applications and is validated through simulations and experiments. Power line faults significantly impact power system security, and making accurate fault diagnosis is essential. Reference [8] uses deep learning models, particularly CNN, to diagnose power line faults, revealing that CNN achieves the highest accuracy and stability with near-zero standard deviation. Recurrent neural networks follow with low accuracy, while Deep belief networks perform with the lowest accuracy with higher variability.

Traditional transmission line fault diagnosis methods often struggle to capture the temporal characteristics of complex faults due to limitations in feature extraction and reliance on heavy preprocessing. To address these challenges, this work proposes a lightweight 1D convolutional neural network (CNN) that directly processes raw time-series data, eliminating the need for handcrafted features or dimensionality reduction techniques. By employing non-uniform kernel sizes across layers, the model captures both localized transient variations and long-duration distortions in fault signals. The use of ReLU activation ensures stable gradient flow and efficient learning without requiring attention mechanisms or residual blocks. Experimental results confirm the model's ability to effectively detect and classify faults in transmission lines using simulated datasets.

The work in [11] proposes a fault recognition method for transmission lines using CNNs combined with continuous wavelet transform (CWT) and independent component analysis (ICA) preprocessing. Relying on voltage signals with moderate sampling frequency, the approach effectively distinguishes faults from disturbances like load and capacitor switching, achieving up to 99% accuracy across different power systems. CNN with the CWT method demonstrates high accuracy. However, it relies on ICA preprocessing, which can introduce computational overhead and complexity; using only voltage signals may limit performance in different cases where current signals provide complementary diagnostic insights. In [12], single-phase-to-ground fault detection in distribution networks using gramian angular field and improved convolutional neural networks, not specifically for transmission lines, was used [13]. LSTM autoencoders for transmission line protection have been deployed in [14]. Thus, it is the need of time to implement the feature extraction methods without much prior knowledge [15], [16].

Usually, CNNs are used in the context of 2-dimensional image settings to extract the spatially temporal features for object detection and voice recognition from videos [17]. However, in a transmission line fault, the post-fault voltage and signal depend on the time in seconds or milliseconds

that the transition occurs in the signal. Therefore, this kind of data we are getting for the post-fault voltage and the current signal is one-dimensional time series data. The anomalies in the 2-D signal image can be extracted and used for fault detection with CNNs; however, the accuracy is limited [18]. The features from the data become a 1-D using filters. CNN with time series data would be able to extract the complex features without missing the critical features that are more indicative of the anomalies in the post-fault current and voltage time series values. While considering the applications of CNN [14], [15], [16], time series-based CNN has been proposed for fault detection and classification in this work. The main contributions of this work are as follows:

II. CONTRIBUTION TO LITERATURE

This work proposes a novel 1D convolutional neural network (CNN)-based architecture for the detection and classification of faults in power transmission lines. The model is designed to directly process raw time-series data comprising voltage and current measurements without the need for extensive feature engineering. The main contribution of this work is as follow

1. This work introduces a 1D CNN model that extracts features directly from raw time-series inputs. This approach preserves the temporal integrity of the signal and reduces preprocessing overhead.
2. Feature extraction using non-uniform kernels by incorporating varying kernel sizes across convolutional layers (kernel size = 3 and 5), the model performs multi-scale feature extraction, allowing it to capture both transient localized variations and persistent fault-induced distortions effectively.
3. The use of ReLU activation functions architecture improves gradient propagation, stabilizes training, and enhances the robustness of the model addressing common issues such as vanishing gradients and slow convergence in deeper neural networks.
4. Flexible adaptation to different signal types and grid conditions. The architecture supports input from multiple signal types (voltage and current) and adapts well to various fault scenarios, making it scalable and suitable for real-world deployment in smart grid environments.

III. TIME SERIES-BASED LEARNING WITH CONVOLUTIONAL NEURAL NETWORK

A. SYSTEM AND FAULT SIMULATION

The system, which is not very complex and is yet complex enough to produce results near a real-time system, has been used, which is the IEEE 9-bus system. It has three generators with generating capacities of 72, 163, and 85 megawatts (MW). Nine buses and transmission lines. Asymmetrical faults are introduced in the transmission line between bus 7 and 8 as shown in Fig. 1 for generating post fault signal data. The details of the IEEE 9-bus system is given in Fig. 2.

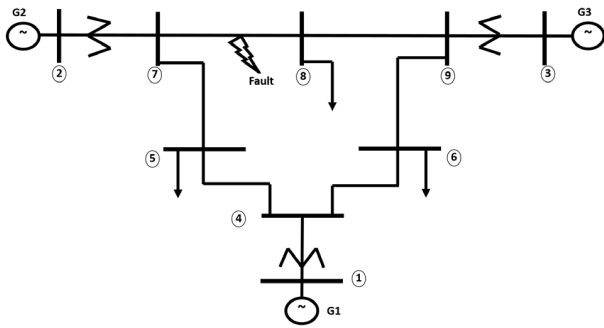


FIGURE 1. IEEE 9-bus system [2].

TABLE 1. System parameter for simulation of faults.

System parameters	Values
fault distance(km)	25,50,75,100
Fault type	A-G, B-G, C-G, AB, BC, CA, AB-G, BC-G, CA-G no-Fault
Features	Post fault Voltage, Post fault current
Frequency (Hz)	60
Fault resistance(ohms)	0.001
Switching time(sec)	0-0.04

The IEEE 9-bus system is simulated in MATLAB/Simulink [19] environment. The parameters for simulating faults in the IEEE 9-bus system is given in Table 1, where the parameters and the values that have been selected for fault cases are displayed. The flow analysis details of all the buses in IEEE 9-bus system are given in Table 2.

Asymmetrical faults have been studied and simulated in this work. Asymmetrical faults are the most common faults in the power system and cause interruptions in the power supply [20]. A fault was generated between bus numbers 7 and 8, and data was collected from bus number 7. As shown in the simulated IEEE 9-bus system in Fig. 2.

The first fault case is a line-to-ground fault (L-G), which includes Phase A to the ground (A-G), Phase B to the ground (B-G), and Phase C to the ground (C-G). The second fault case is double fault, which includes Phase A to Phase B, Phase B to C, and Phase C to A fault. The third case is double line to ground fault, which includes phases AB to ground (AB-G), Phase BC to ground (BC-G), and Phase CA to ground fault (CA-G). And the last case is No-fault. After generating faults between seven and eight buses, the post-fault voltage and current time series data for all the cases were collected on bus number seven and a database.

The three-phase fault block is shown in Fig. 3. It shows the parameters that have been selected to generate and simulate the three-phase faults in the system.

B. DATA ACQUISITION

Asymmetrical faults are generated in the transmission line between buses 7 and 8. By changing the various system

parameters, post-fault voltage and current data samples for all fault types are generated. The system parameters are mentioned in the Table 1. The three-phase fault and its parameters are given in Fig. 3. One of the tunable parameters in this paper is the distance at which the fault is generated, the distance as mentioned in Table 1 is 25, 50, 75, and 100km. At each distance, all cases are simulated to get the time series data.

Fig 5. shows all the fault cases that have been considered and collected as data samples for developing a model for fault detection and classification. These include phase A-G, B-G, C-G, AB-G, BC-G, CA-G, AB, BC and CA faults. After updating the system parameters and three-phase fault parameters, post-fault voltage and current data samples at each location are sent to the workspace. The same process is repeated for all the fault cases and no-fault case to collect the data samples. The sample data is given in Table 3 for fault detection and Table 4 shows the sample data for fault classification.

In Table 3 and Table 4 the six features of post fault that is current (I_a , I_b , I_c), and voltage (V_a , V_b , V_c) are used as input data during training. The last column in Table 3 and Table 4 are labels for detection and classification these are used as output data during training. The difference between fault detection and fault classification data is the labels we added in the last column of each data set. For example, in the fault detection data sample, the label for fault cases is 1, and the label for non-fault cases is zero. In this work, supervised learning has been used. Therefore, these labels will help classify the fault and no-fault cases accurately during training and testing. In the case of fault classification, each fault case is labeled with a different label. For example, A-G = 1, B-G = 2, C-G = 3, AB = 4, BC = 5, CA = 6, AB-G = 7, BC-G = 8, CA-G = 9, No-Fault = 0. These labels in each fault classification will help learn the behavior of input features and output labels.

The total data samples collected for all nine fault cases and No-fault cases is 3240. The sampling time, as shown in Fig. 4, for each data sample, is 0.0005 seconds, and the total simulation time is 0.04 seconds. Three-phase voltage and current samples act as input data points, and the labels act as output in the fully connected layer in the model of CNN-based fault detection and classification [17]. After selecting the parameters mentioned in the workspace block in Fig. 4, the data is sent to the workspace and generates a data store for voltage and current. The total data was later divided into two sets: a training set and a testing set. The training set, which is 60 percent, is used for training to develop CNN trained model, and the testing data set, which is 20 percent, is used to test the model for validating the efficiency of the CNN-trained model.

IV. CONVOLUTIONAL NEURAL NETWORK-BASED FAULT DETECTION AND CLASSIFICATION

Among deep neural networks, CNNs are the most famous among all the networks [21] due to their wide range of applications, including object detection, audio recognition, aircraft sensor faults, etc [22].

TABLE 2. IEEE 9 bus-system flow chart [19].

Block Name	Block Type	Bus Type	Bus ID	Vbase (kV)	Vref (pu)	Vangle (deg)	P (MW)	Q (Mvar)	Qmin (Mvar)	Qmax (Mvar)	V_LF (pu)	Vangle_LF (deg)	P_LF (MW)	Q_LF (MVA)
247.5 MVA, 16.5 kV	Vsrc	swing	BUS_1	16.5	1.04	0	0	0	inf	Inf	1.04	0	72.0099	27.6543
Load Flow Bus1	Bus	-	BUS_4	230	1	0	0	0	0	0	1.0256	-2.2219	0	0
125 MW 50 MVAR/Three-Phase Parallel RLC Load	RLC load	PQ	BUS_5	230	1	0	125	50	inf	Inf	0.9952	-4.1438	125	50
90 MW 30 MVAR/Three-Phase Parallel RLC Load	RLC load	PQ	BUS_6	230	1	0	90	30	inf	Inf	1.0126	-3.5293	90	30
Load Flow Bus4	Bus	-	BUS_7	230	1	0	0	0	0	0	1.0234	3.2304	0	0
Load Flow Bus5	Bus	-	BUS_9	230	1	0	0	0	0	0	1.0326	2.3507	0	0
100 MW 35 MVAR/Three-Phase Parallel RLC Load	RLC load	PQ	BUS_8	230	1	0	100	35	inf	Inf	1.0168	1.301	100	35
192 MVA, 18 kV	Vsrc	PV	BUS_2	18	1.025	0	163	0	inf	Inf	1.025	8.7966	163	10.8154
128 MVA, 13.8 kV	Vsrc	PV	BUS_3	13.8	1.025	0	85	0	inf	Inf	1.025	5.0415	85	11.0624

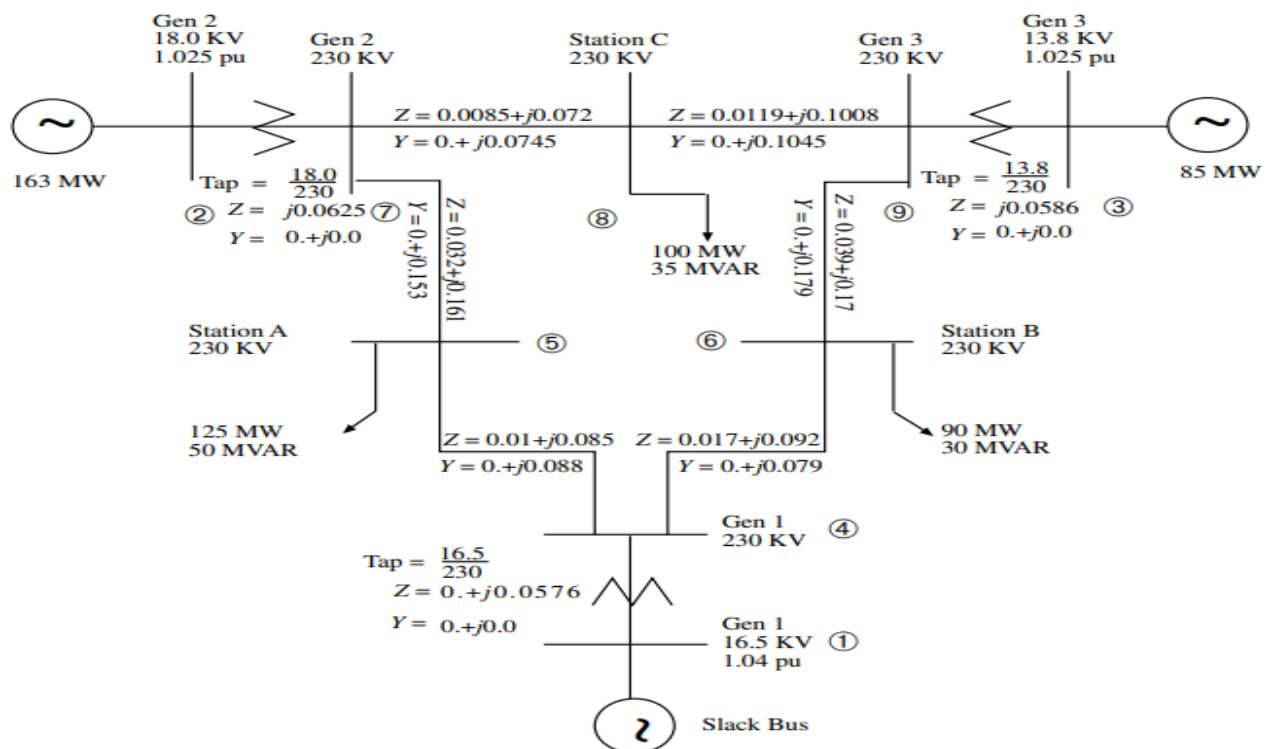


FIGURE 2. IEEE 9-bus simulation in MATLAB, Simulink [2], [19].

CNNs consist of two main parts [7]: feature extraction and classification; feature extraction is done by using convolutional layer, followed by an activation function layer to generate a deep neural network [23]. We have introduced a very simple network, which is effective in feature extraction, simple in design, accurate in fault detection and less computation burden to train. This fault detection and classification network flow is shown in Fig. 6.

The proposed CNN consists of 11 layers, including three 1-D convolutional layers with 16 and 32 filters, followed by ReLU activations. Each convolutional block is connected

to a fully connected layer for feature interpretation [5]. The network ends with a regression output layer, predicting a continuous value based on the input signal.

Each convolutional layer has 16 and 32 filters of size 3 and 5, enabling the network to capture localized signal that changes effectively, which is important for identifying transient faults or anomalies in power system signals [24]. In one-dimensional signal processing, the convolutional kernel plays a vital role in extracting features by sliding across the time axis of the input signal. Each kernel acts as a trainable filter that captures specific temporal patterns or transitions

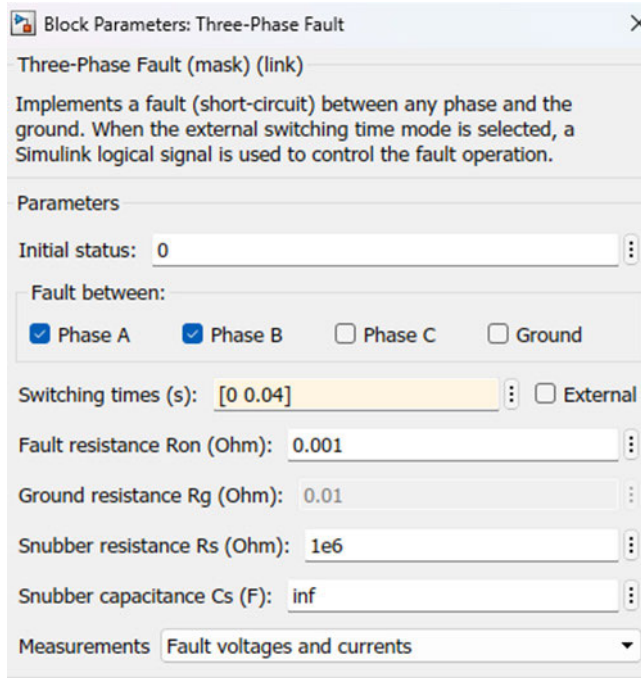


FIGURE 3. Three-phase fault parameters [19].

TABLE 3. Data sample for fault detection.

Fault cases	Ia	Ib	Ic	Va	Vb	Vc	detection
A-G	-0.279	-0.2472	0.5262	-0.4447	-0.4092	0.8538	1
	-0.6848	-0.3048	0.5453	-0.3001	-0.5424	0.8425	1
	-0.298	-0.4012	0.4957	-0.145	-0.6564	0.8014	1
B-G	-0.074	-1.0695	0.517	-0.0271	-0.0021	0.9243	1
	0.0716	-1.6003	0.5221	0.0356	-0.3905	0.7555	2
	0.1092	-2.2096	0.4031	0.2008	-0.1296	0.6639	1
C-G	0.2452	-0.4734	1.9431	0.2134	-0.9704	-0.2852	1
	0.2942	-0.5168	2.0563	0.601	-0.7309	0.4004	1
	0.3932	-0.4816	2.3837	0.7365	-0.6964	-0.0716	1
AB	1.7019	-1.9427	0.2407	-0.0956	-0.2883	0.3838	1
	2.1242	-2.273	0.1487	0.0267	-0.2596	0.2327	1
	2.5178	-2.5693	0.0514	0.0476	-0.1214	0.0738	1
BC	0.1972	-2.321	2.1237	0.3313	-0.0677	-0.2636	1
	0.2852	-2.7609	2.4756	0.4741	-0.5803	0.1063	1
	0.3631	-3.1101	2.7469	0.6	-0.1694	-0.4306	1
CA	1.9674	-0.0074	-1.9599	0.0393	-0.0019	-0.0374	1
	1.8202	0.0914	-1.9116	-0.1384	0.1592	-0.0208	1
	2.1608	0.187	-2.3477	0.1432	0.3146	-0.4578	1
AB-G	-0.6187	-0.4136	0.5499	-0.0953	-0.1947	0.8541	1
	-0.6332	-0.8301	0.5195	-0.0164	-0.1479	0.8201	1
	-0.4219	-1.1849	0.4839	0.0911	-0.1063	0.7895	1
BC-G	0.0803	-1.6055	2.1023	0.1462	-0.0671	0.0548	1
	0.1857	-2.0568	2.3864	0.3048	-0.2175	0.1266	1
	0.279	-2.4979	2.4906	0.4366	-0.1086	-0.0804	1
CA-G	0.6962	-0.515	1.9526	0.0958	-0.799	0.0931	1
	1.0915	-0.4674	1.7644	0.2036	-0.7369	-0.1129	1
	1.6022	-0.4082	1.3821	0.1539	-0.6295	-0.033	1
No-fault	0.5018	-0.3888	-0.113	0.8223	-0.6278	-0.1945	0
	0.5228	-0.3154	-0.2074	0.8546	-0.5067	-0.3479	0
	0.5252	-0.2308	-0.2944	0.8566	-0.3676	-0.489	0

TABLE 4. Data sample for fault classification.

Fault cases	Ia	Ib	Ic	Va	Vb	Vc	Classification
A-G	-0.279	-0.2472	0.5262	-0.4447	-0.4092	0.8538	1
	-0.6848	-0.3048	0.5453	-0.3001	-0.5424	0.8425	1
	-0.298	-0.4012	0.4957	-0.145	-0.6564	0.8014	1
B-G	-0.074	-1.0695	0.517	-0.0271	-0.0021	0.9243	2
	0.0716	-1.6003	0.5221	0.0356	-0.3905	0.7555	2
	0.1092	-2.2096	0.4031	0.2008	-0.1296	0.6639	2
C-G	0.2452	-0.4734	1.9431	0.2134	-0.9704	-0.2852	3
	0.2942	-0.5168	2.0563	0.601	-0.7309	0.4004	3
	0.3932	-0.4816	2.3837	0.7365	-0.6964	-0.0716	3
AB	1.7019	-1.9427	0.2407	-0.0956	-0.2883	0.3838	4
	2.1242	-2.273	0.1487	0.0267	-0.2596	0.2329	4
	2.5178	-2.5693	0.0514	0.0476	-0.1214	0.0738	4
BC	0.1972	-2.321	2.1237	0.3313	-0.0677	-0.2636	5
	0.2852	-2.7609	2.4756	0.4741	-0.5803	0.1063	5
	0.3631	-3.1101	2.7469	0.6	-0.1694	-0.4306	5
CA	1.9674	-0.0074	-1.9599	0.0393	-0.0019	-0.0374	6
	1.8202	0.0914	-1.9116	-0.1384	0.1592	-0.0208	6
	2.1608	0.187	-2.3477	0.1432	0.3146	-0.4578	6
AB-G	-0.6187	-0.4136	0.5499	-0.0953	-0.1947	0.8541	7
	-0.6332	-0.8301	0.5195	-0.0164	-0.1479	0.8201	7
	-0.4219	-1.1849	0.4839	0.0911	-0.1063	0.7895	7
BC-G	0.0803	-1.6055	2.1023	0.1462	-0.0671	0.0548	8
	0.1857	-2.0568	2.3864	0.3048	-0.2175	0.1266	8
	0.279	-2.4979	2.4906	0.4366	-0.1086	-0.0804	8
CA-G	0.6962	-0.515	1.9526	0.0958	-0.799	0.0931	9
	1.0915	-0.4674	1.7644	0.2036	-0.7369	-0.1129	9
	1.6022	-0.4082	1.3821	0.1539	-0.6295	-0.033	9
No-fault	0.5018	-0.3888	-0.113	0.8223	-0.6278	-0.1945	0
	0.5228	-0.3154	-0.2074	0.8546	-0.5067	-0.3479	0
	0.5252	-0.2308	-0.2944	0.8566	-0.3676	-0.489	0

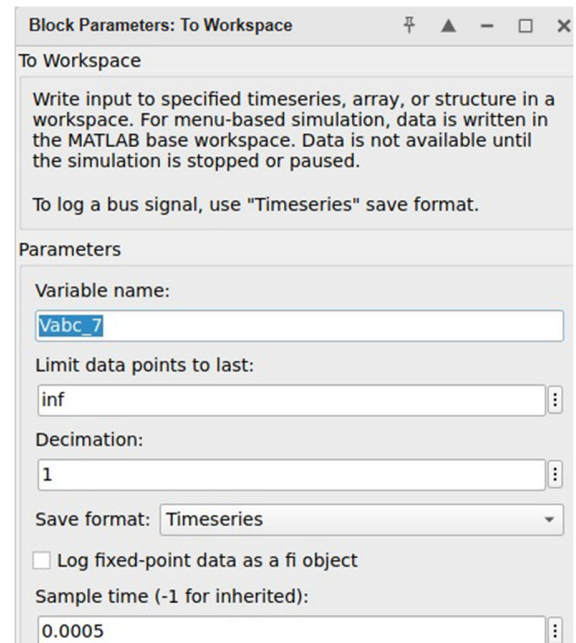


FIGURE 4. Time series data sampling time [19].

in the signal, such as sudden spikes, oscillations, or gradual shifts common in fault scenarios. Unlike 2D convolutions that detect spatial patterns in images, 1D kernels focus purely

on the temporal dynamics of multivariate signals. In the proposed architecture, the input data consists of 3240 time steps and 6 feature voltage and current measurements, where each 1D convolution is applied across time while aggregating information across features [25].

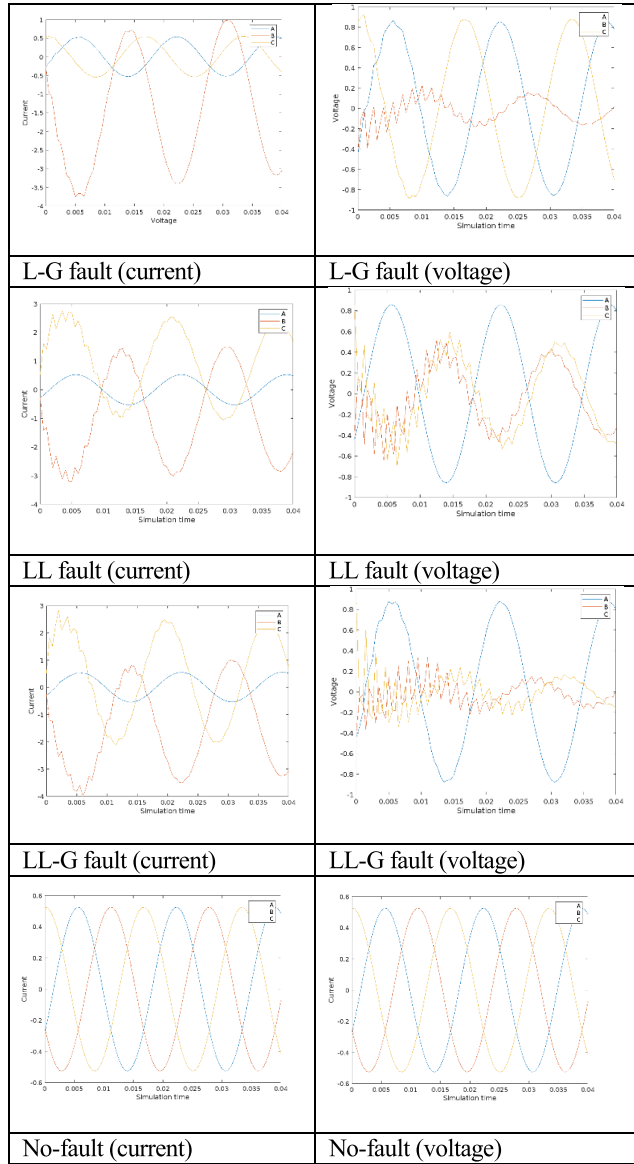


FIGURE 5. Fault and no-fault cases [18], [19].

ReLU is used after each convolutional layer, introducing non-linearity and helping the model learn complex patterns [26] while keeping the computation simple and efficient. Additionally, ReLU's gradient is simple and non-saturating for positive inputs, effectively mitigating the vanishing gradient problem commonly observed in deeper networks [27]. This results in better gradient flow during backpropagation and faster convergence during training.

The use of a regression output layer instead of classification suggests the model predicts a continuous value, possibly the severity or likelihood of a fault. This is more informative than a binary result and allows for threshold-based fault decisions. This approach allows the model to perform multi-class fault classification without the need for a conventional softmax layer. The use of regression output simplifies the network architecture while maintaining compatibility

with fault index prediction, which is particularly advantageous for real-time protection and monitoring applications in power systems. The proposed method has been empirically validated, demonstrating high classification accuracy and reliable fault discrimination when applied to time-series voltage and current signal data.

Each layer has a manageable number of learnable parameters, which helps maintain a balance between performance and generalization, which is, essential when working with datasets that may be limited in size or diversity.

The model is implemented using MATLAB Deep Learning Toolbox, a reliable platform for signal processing and engineering applications. The use of the Adam optimizer ensures fast and adaptive learning with minimal manual tuning.

A. FAULT DETECTION AND CLASSIFICATION TRAINING AND TESTING

For the training and testing with time series data, the CNN network is updated with the options like filter activation functions. There are six features, and one output for fault detection and has two classes, namely fault and No-fault class which has been represented by 1 and 0. In the fault classification, there are 10 classes, each representing the fault cases such as AG, BG, CG, AB, BC, CA, AB-G, BC-G, and CA-G fault, and each represents the number.

At the convolutional layer, input data is convolved with the number of filters with a specific dimension defined for the particular convolutional layer. This layer is followed by a rectified linear layer (ReLU) to define a feature version from the feature map. The operation of the convolutional layer and ReLU takes place using the equation (1) [21], [23], [28].

$$v_{l,j}^x = f \left(\sum_m \sum_{h=0} k_{l,j,m}^h v_{(l-1),m}^{x+h} \right) + b_{l,j} \quad (1)$$

where $v_{l,j}^x$ is the feature at x position on the vector of j th in the l th convolutional layer, $b_{l,j}$ is the bias of the k th feature vector in the l th layer, m indicates the feature vector index in the $(l-1)$ th layer which is the current feature vector. $k_{l,j,m}^h$ is the value of kernel at h th position of the m th feature vector at $(l-1)$ th layer. The length of the kernel is represented by H_l . F is the activation function. This is how Feature mapping and feature vectors develop at convolutional and ReLU layers [7].

The filters used in a convolutional layer in the proposed models are 16 and 32 of size 3 and 5. The convolutional kernel operation in the 1-D domain is central to this model. These kernels slide across time steps, performing localized filtering to extract patterns such as abrupt signal deviations, oscillations, or distortions typical of fault events. By learning these patterns from data, the model develops sensitivity to both transient localized variations and persistent fault-induced distortions, making it effective across a variety of fault types and durations.

The next layer is a fully connected layer, which simplifies the outputs from previous layers. It does that by reducing the dimensionality of features to a manageable level.

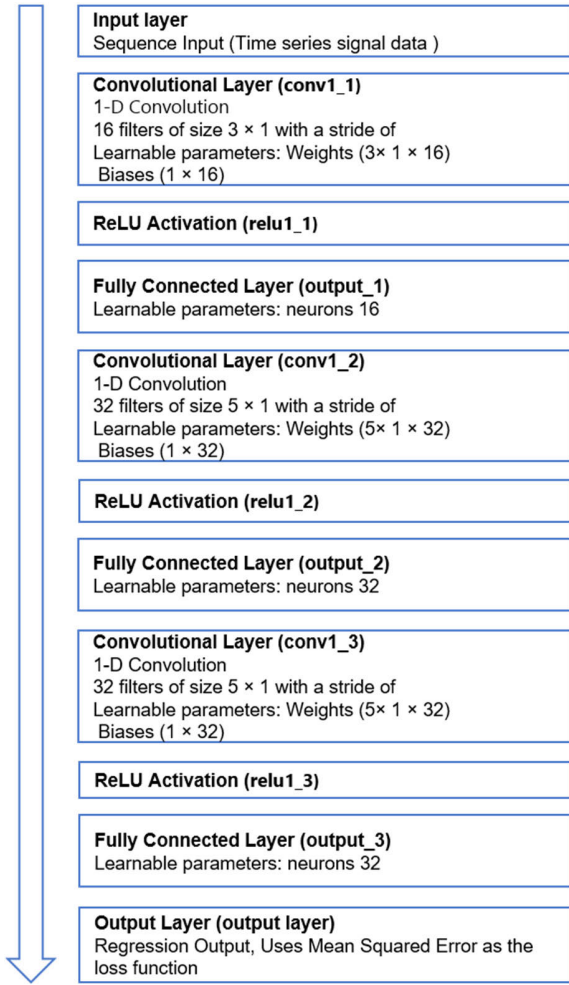


FIGURE 6. Fault detection and classification proposed train network flow with 11 layers [19].

For fault detection and fault classification, we repeated these three layers and took the output from the final layer, which in this proposed model is the regression layer.

This is usually not the standard way of CNNs; in standard CNNs, the last layers are always fully connected: the softmax layer and classification layer [6]. We used the combination of convolutional, ReLU, and fully connected to get the benefit of deeper feature extraction [29], which is done by convolutional layers.

The repetition of layers makes the model able to learn hierarchical patterns. The higher layer focuses more on global patterns, and the lower layer focuses on smaller details. Repetition is good for learning more complex behaviors between input and output. Multiple ReLU layers can enhance nonlinearity, and it reduces the whole computation time [30]. The last layer is the regression layer, which computes the continuous values, and it is suitable for fault detection and fault classification as we represent the fault cases with different values. After updating all the options and layers for the model, we train the model with a training set of fault data. After training, we test the trained model with the testing

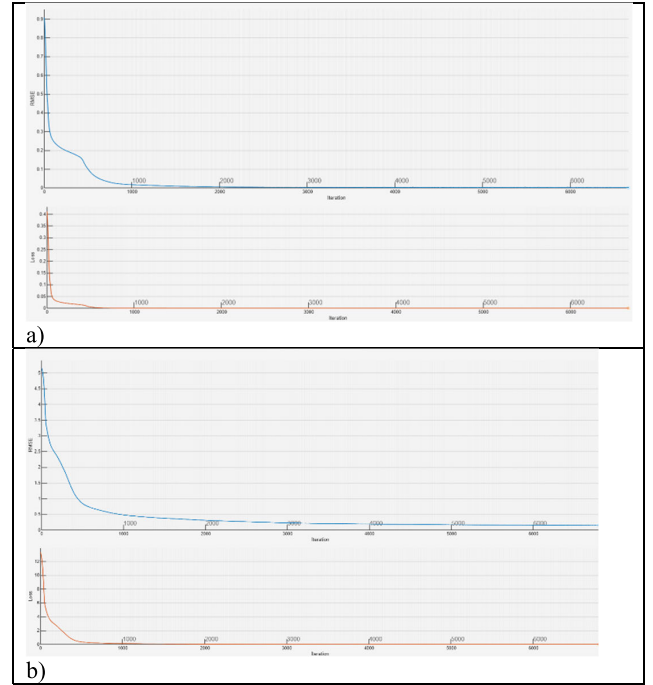


FIGURE 7. Training progress for fault a) detection b) classification.

data set. The training, testing progress, and the loss are captured and displayed here in Fig. 7. For the evaluation of the trained model for detection the fault and classification the faults we used, RMSE, MAE and R^2 the equations are given below:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (Output_{predicted} - Output_{actual})^2} \quad (2)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (Output_{predicted} - Output_{actual})^2}{\sum_{i=1}^n (Output_{actual} - output_{mean})^2} \quad (3)$$

$$MAE = \sum_{i=1}^n \frac{|Output_{predicted} - Output_{actual}|}{n} \quad (4)$$

where RMSE is root mean square error value, “n” is the number of fault samples, $Output_{predicted}$ is out generated by the trained model, $Output_{actual}$ is the actual output in fault detection case fault = 1 and no-fault = 0 are the actual outputs. R^2 is coefficient of determination and MAE stands for mean absolute error.

V. EXPERIMENTAL RESULTS

To check the effectiveness of the model, we tested the Fault detection and fault classification model with the IEEE 9 bus system. Different faults have been generated at different locations and tested using the fault detection CNN model. The results for fault detection for different faults are displayed in Fig. 8. where the fault cases AG, BG, CG, AB, BC, CA, AB-G, BC-G, and CA-G faults, as shown in Fig. 8 and Fig. 9, have been generated for the half cycle so the case of the fault happened from 0.01 to 0.02 seconds rest of the time there is no fault. The CNN base fault detector detected the fault and

updated it as one where there was a fault, and where there was no fault, it updated as zero. Mean square error and mean absolute error have been calculated for the effectiveness of training and testing results and are displayed in the Table 4, where it shows that the testing mean square error (MSE) with the test data set we achieved is 1.4410×10^{-4} for fault detection, which means there is nearly zero error in the fault detection. CNN-based fault detectors can detect the fault with precision. As this is also displayed in Fig. 8, each fault and no-fault condition is precisely detected and shown as zero and one in the predicted graph for each fault case.

The predicted graph for all the fault detection cases in Fig. 8 can be analyzed as follows: in the case of AG fault, there is accurate detection with a curve aligned well with the visual onset and offset of the fault. In the case of the BG fault model, detection is precise and tightly aligned with the fault. Similarly, the CG fault model well-detected a fault with no delay. Similarly, for double line faults, all faults were detected very well and at the right time. In the case

of double line-to-ground faults, the model performed very well and captured the complex faults involving the ground.

TABLE 5. Training and testing results.

	model	MSE	RMSE	MAE	R ²
Train	Detection	2.0721×10^{-7}	4.5521×10^{-4}	1.2466×10^{-4}	1.0000
	Classification	0.0149	0.1220	0.0891	0.9982
Test	Detection	1.4410×10^{-4}	0.0120	0.0018	0.9987
	Classification	0.1442	0.3798	0.3019	0.9806

In the case of fault classification, the MSE is 0.1442. The predicted results for each fault case generated by the CNN-based fault classifier are displayed in Fig. 9. In cases of a CNN-based classifier, some faults are classified correctly, but some cases are classified as different faults. For example, the fault AG in the predicted graph in Fig. 9, fault AG is generated from 0.01 to 0.02 seconds rest of the time is a normal condition; in this case, the fault is classified as one,

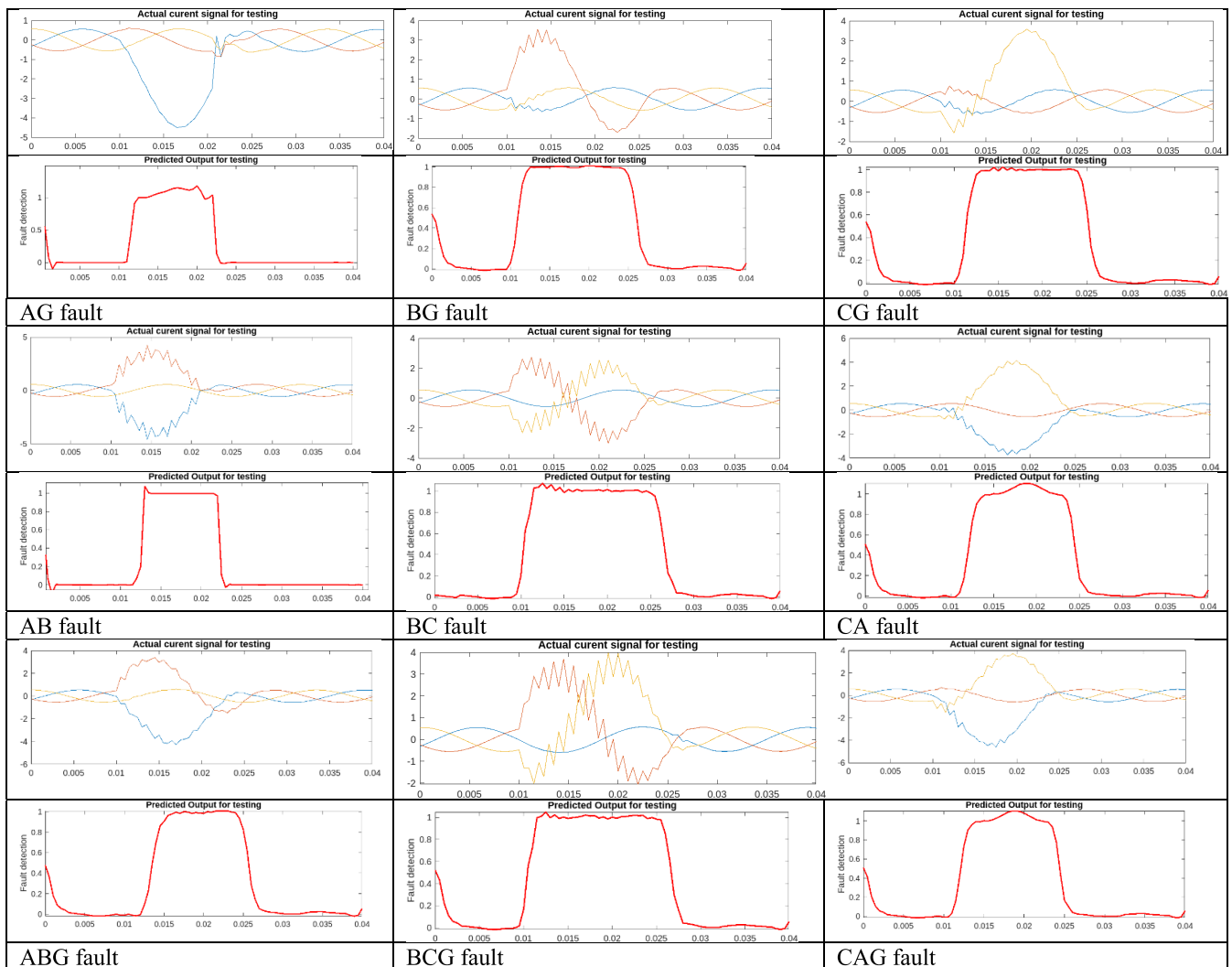


FIGURE 8. CNN-based fault detector response for fault cases [19].

which means it is AG fault. In the case of fault BG, which is supposed to be predicted as two, it is shown in the predicted graph of BG fault that the classifier predicted as almost 2. In the case of fault CG, the predicted graph shows it as 3. The classifier classified it correctly. Next, in fault cases such as AB fault, the classifier is supposed to predict as, and it predicted nearly four. The fault BC is also predicted as four, which is incorrect. CA fault is classified as 6, which is correct. In the case of double line to ground faults, AB-G is predicted correctly, and the other two faults, BC-G and CA-G fault faults, were classified as other faults. Therefore, the performance of the fault-based classifier is good but not ideal. The Mean square error for training and testing of the CNN-based classifier is given on Table 5.

The performance of the classifier can be enhanced by adding more depth to the network, which is possible by adding more layers as convolutional layers and increasing the weights in the fully connected layers. The confusion matrix has also been generated to support the evolution

of CNN-based models. The results of the confusion matrix for testing data for CNN-based fault detector and CNN-based fault classifier are shown in Fig. 10. From the confusion matrix, the testing data that have been used to detect the fault cases were a total of 648, 81 were fault cases, and 567 No-fault cases.

Fig. 8 and Fig. 9. shows all the cases that have been detected correctly and incorrectly for fault and No-fault cases; in the case of fault detection, all the faults and No-fault cases have been detected correctly. In the case of classification, a few cases have been classified as other faults, as shown in the confusion matrix. In the case of fault detection, there are only two classes, so it is easier for the proposed technique to understand the behavior. In the case of fault classification, it missed some faults and classified some faults as another fault; it shows that if there are more additional conditions, there would be more distinguished features to differentiate more precisely and more depth in the network [31] would be able to distinguish without missing any case.

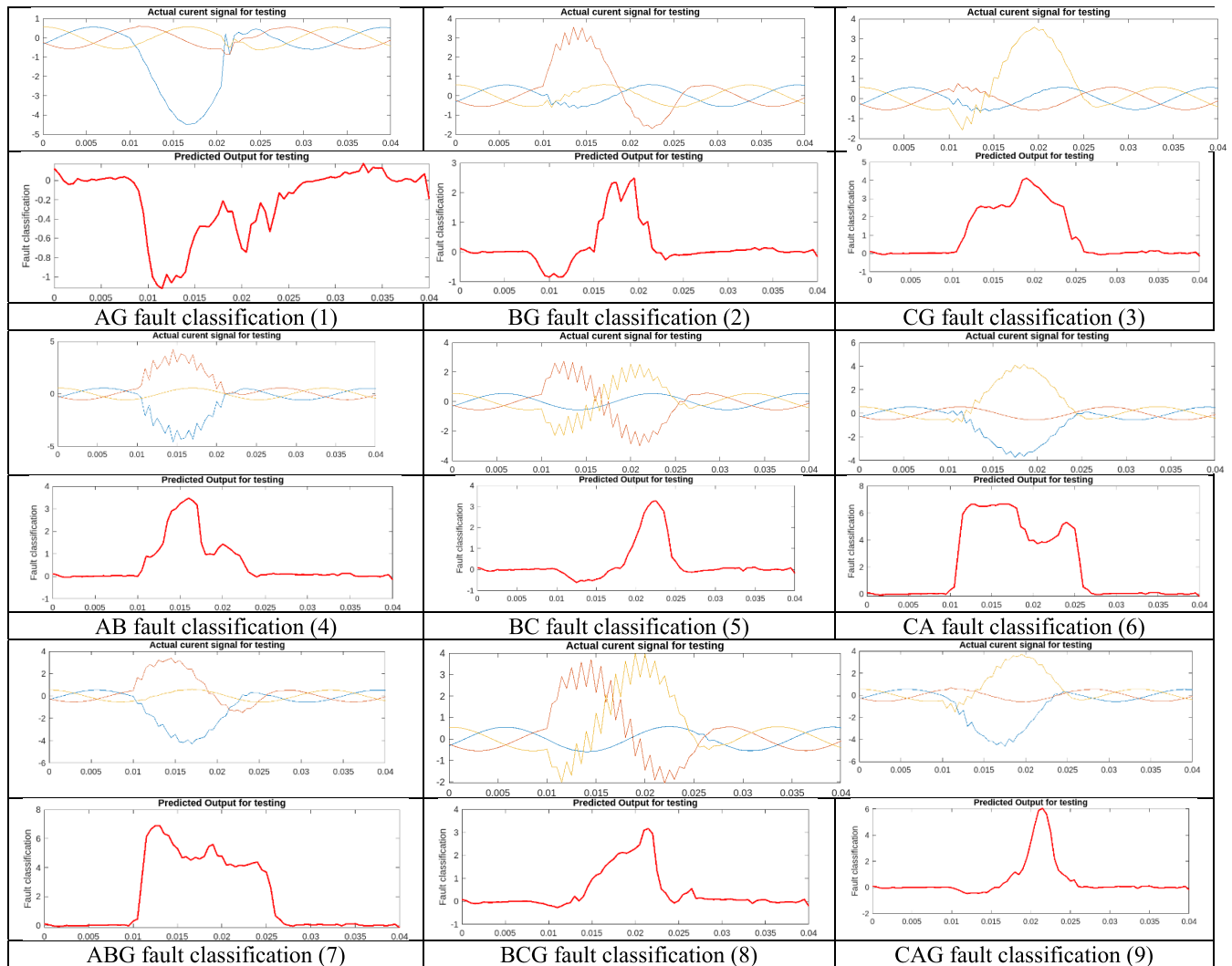


FIGURE 9. CNN-based fault classifier response for fault cases [19].

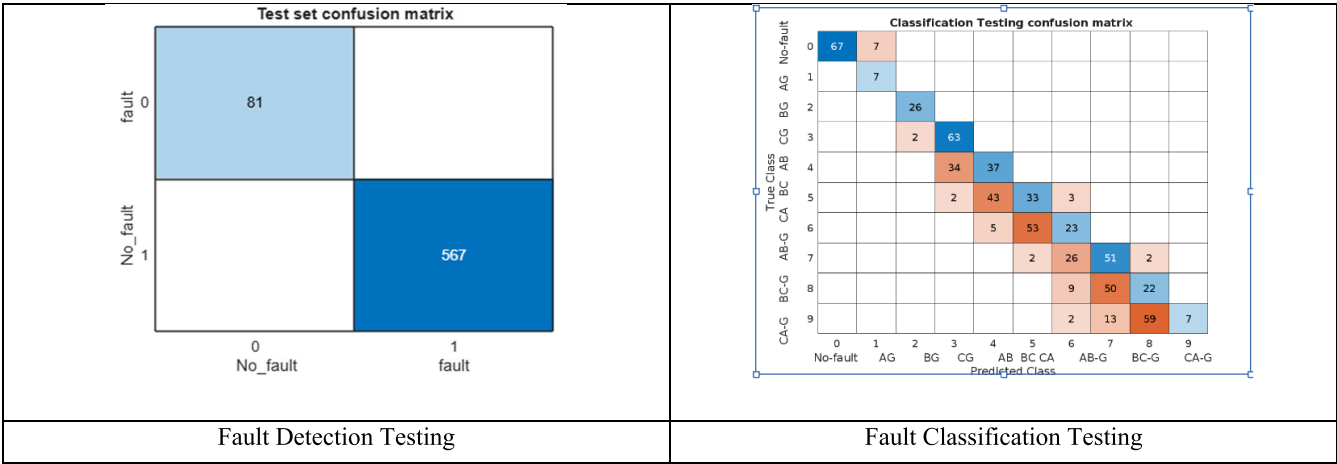


FIGURE 10. Confusion matrix for fault detection and classification [19].

The proposed deep learning techniques are compared with different CNN techniques applied in fault detection and classification in the Table 6. The proposed fault detection model is simple, yet its percentage efficiency for fault detection is 100%, as shown in Fig. 10, which is comparatively better and more efficient for fault detection. Deep learning-based fault detection in multi-phase permanent magnet synchronous motor (PMSM) drive systems for four-wheel-independent-drive electric vehicles (4WID-EVs) for three-channel samples has an accuracy of 96.10%. The method in [32] diagnoses faults for rolling bearings in electric motors, addressing challenges from noise and nonstationary vibration signals. It integrates improved variation mode decomposition (VMD) with a CNN-BiLSTM network. The Osprey-Cauchy-Sparrow search algorithm (OCSSA) optimizes VMD parameters for efficient signal decomposition based on minimum envelope entropy. The accuracy is 100% due to the hybrid nature of the method. The proposed method is able to achieve the same accuracy as a non-hybrid method, which reduces the training and testing time comparatively. A multibranch and multiscale CNN for fault diagnosis of wheelset bearings in high-speed trains is proposed [33], with 12 different faults studied. The proposed method for fault detection is able to detect all faults. The CNN-based proposed method is simple, not deep, and the results generated for fault detection are limited to the IEEE 9 bus system. This proposed system needs to be tested with a complex system for its validation.

The proposed 1D CNN model demonstrates high accuracy and efficiency in fault detection and classification using raw time-series data. However, its current evaluation on simulated data from the IEEE 9-bus system limits its generalizability to real-world applications and to larger networks. Simulated signals may not capture the noise, variations, and complexities of actual power system conditions. Moreover, the model’s performance may vary with different network topologies and transmission configurations, which were not

explored in this study. As system size increases, so does the complexity of fault scenarios and volume of time-series data, potentially affecting training time, memory usage, and classification accuracy, especially in the presence of class imbalance or sparse fault types.

To adapt this method for larger, real-world systems, several improvements are necessary. These include using zone-based or modular training, enhancing feature extraction to manage input dimensionality, and exploring more scalable architectures like LSTM with attention or transformer-based models to better capture temporal dependencies. Additionally, preparing a well-balanced and diverse dataset will be critical for robust generalization and accurate real-time deployment, particularly on edge devices where further model optimization may be needed.

TABLE 6. Comparison of CNN-based techniques with proposed techniques.

Method	Sample	Accuracy(%)
OCSSA-VMD-CNN-BiLSTM. [32]	Vibration signal	100
Multiscale CNN [33]	Vibration Data	93.97
1-D CNN [34]	Three channel signal	96.10
CNN-LSTM [35]	Power Quality Disturbances	98.89
(Proposed) CNN	Three-phase fault signal	100

VI. CONCLUSION

This paper proposes a simple yet effective one-dimensional convolutional neural network model for detecting and classifying faults in electrical transmission lines using raw time-series data. The model eliminates the need for manual feature engineering by automatically extracting features through stacked convolutional layers. These layers use multiple filters with varying kernel sizes to capture both localized transient changes and longer-term fault-induced distortions

in the signal. Non-linear activation functions, specifically the rectified linear unit, are applied after each convolution to introduce non-linearity, enhance learning capacity, and prevent vanishing gradient issues. This structure enables the model to learn complex fault patterns from multichannel time-series inputs, such as voltage and current measurements, with high accuracy and minimal computational complexity. The results deduced in this work for fault detection and fault classification represent the potential of CNN-based models for fault detection and fault classification. These models have been tested with the IEEE 9-bus system, and they can be tested with a real-time system for the effectiveness of these techniques and models. The presented CNN-based fault detection model can detect the faults with 100 percent precision; however, for fault classification, it needs a deeper network. This technique could be used in the future to find fault locations in power systems.

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SOMCHAT JIRIWIBHAKORN (Senior Member, IEEE) received the B.Sc. and M.Sc. degrees in electrical engineering from the King Mongkut's Institute of Technology Ladkrabang (KMITL), Bangkok, Thailand, in 1994 and 1997, respectively, and the Ph.D. degree in electrical engineering from Imperial College London, U.K., in 2000.

He has been an Associate Professor with the Department of Electrical Engineering, School of Engineering, KMITL, since 2006. His research

interests include power system stability, power system optimization, power system planning and forecasting, and applications of neural networks and ANFIS in power engineering.



SHAZIA KANWAL received the B.S. degree in electronics from the Sir Syed University of Engineering and Technology, Karachi, Pakistan, and the M.Sc. degree in electronics from the King Mongkut's University of Technology, North Bangkok, Thailand. She is currently pursuing the D.Eng. degree in electrical engineering with the King Mongkut's Institute of Technology Ladkrabang (KMITL). Her research interest includes artificial intelligent based fault analysis in transmission lines.

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